Lesson 2.8 Guided Notes

Session 1 · Finding the Mean

I can find the mean of a data set by sharing the total equally among the values. · 6.SP.B.5c

Word / Spanish	Meaning — complete the blank	Example
mean / media	The fair share: add all values, then _____ by how many there are.	(3 + 5 + 6 + 10) ÷ 4 = 6
average / promedio	Another _____ for the mean.	the average score
measure of center / medida de centro	One number that tells a _____ value of the data.	mean, median
data set / conjunto de datos	All the _____ you collected.	8, 15, 11, 6

Key idea

Mean = _____ of the values ÷ _____ of values.

Count every value, even a _____.

The mean lands between the _____ and greatest values.

Worked example

Four friends have 3, 5, 6 and 10 stickers. Find the mean.

Add: 3 + 5 + 6 + 10 = _____

Count the values: there are _____.

Divide: 24 ÷ 4 = _____. Check: 6 is between 3 and 10.

Answer: The mean is _____ stickers.

You try

1. Find the mean of each data set.
a. 4, 8, 6, 10, 7 _____
b. 3, 5, 7 _____

_____

2. Five families paid $18, $24, $15, $21 and $22 for tickets. Find the mean amount.

_____

Sum it up: The mean is a fair share because ___.

_____

Session 2 · Reaching a Target Mean

I can find the value a data set still needs in order to reach a target mean. · 6.SP.B.5c

Word / Spanish	Meaning — complete the blank	Example
target mean / media objetivo	The mean you _____ to reach.	a goal mean of 29
target total / total objetivo	Target mean × number of _____.	8 × 4 = 32
missing value / valor faltante	Target total − the sum you _____ have.	32 − 22 = 10

Key idea

Work _____: target total = target mean × number of values.

Then _____ the sum so far to find the missing value.

Worked example

Quiz scores are 6, 7 and 9. What fourth score gives a mean of 8?

Count all the scores, including the missing one: _____.

Target total: 8 × 4 = _____. Sum so far: 6 + 7 + 9 = 22.

Subtract: 32 − 22 = _____.

Answer: The fourth score is _____. Check: 32 ÷ 4 = 8

You try

1. Fill in each blank.
a. Mean 25, 4 values. Total = _____
b. Need 72, have 57. Missing = _____

_____

2. Maria scored 85, 90 and 88 on three quizzes. What score on quiz 4 gives a mean of 90?

_____

Sum it up: To find a missing value, I first find the ___ because ___.

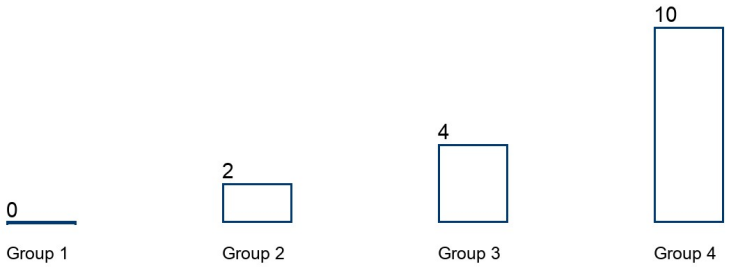
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Session 1 · Model, reason, and explain

Finding the Mean · 6.SP.B.5c

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Four groups collected 0, 2, 4, and 10 cans. Use a fair-share model to find the mean. Explain why the group with zero cans is still included in the divisor.



_____

_____

_____

Extend your reasoning

Must a group actually have the mean number? Construct a two-group counterexample.

_____

_____

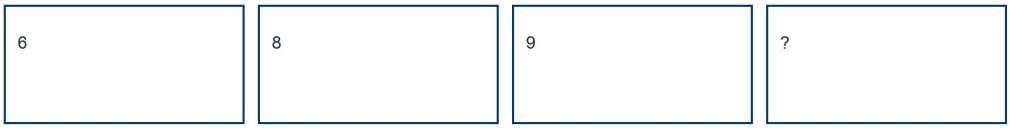
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Reaching a Target Mean · 6.SP.B.5c

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Three quiz scores are 6, 8, and 9. What fourth score gives a mean of 8? Each quiz has a maximum of 10. Could a fourth score produce a mean of 9? Explain.



_____

_____

_____

Extend your reasoning

Find the greatest possible final mean and explain why it is a maximum.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?